01-Apr-2026

Agenda:

Stats Day-2:

- → Data types uses in stats
- → Measurment of central tendency
- → Measurment of dispersion

Types of data (programming):

int, float, string, boolean, list, set, datetime, dataframe → (python, pandas, ..)

Data types of statistics:

Student	Marks	Gender	Grade	Passed	Height	weight
A	85	M	A	Yes		
B	72	F	B	Yes		
C	40	M	D	No		

(1) continous data

data that can take any value in a range (including decimals)

values → 0 – 100 → infinite value

0, 1, 2, 3 .... 100 → 101

0, 0.1, 0.2, ... 1 → 11

0, 0.01, 0.02, 0.03 → 0.1

0, 0.001, 0.002 → infinite precision

Example: Height : 170.236 cm

weight : 65.75 kg

Temperate : 36.6°C

(2) Discrete data

→ countable values (integer only)

Example :

Number of students = 30
No. of questions = 10

? 2.5 students ⟶ ✗

(3) Categorical Data (labels, not numbers)

No order (Nominal)

Example :

Gender : M, F
Colors : Red, Blue

→ Rank ? → F > M , M > F

Has order (ordinal)

Example :

Grade : A > B > C > D

(4) Binary data

only 2 values :

→ Yes / No
→ pass / fail

Nominal Data (Categorical – No Order)
Examples:

Gender: Male, Female, Other
Blood Group: A, B, AB, O
Colors: Red, Blue, Green
Country Names: India, USA, Japan
Type of Car: SUV, Sedan, Hatchback

Ordinal Data (Categorical – Ordered)

Examples:

Satisfaction Level: Poor, Average, Good, Excellent
Education Level: High School, Bachelor's, Master's, PhD
Rankings: 1st, 2nd, 3rd
Pain Level: Mild, Moderate, Severe
Star Ratings: ⭐ to ⭐⭐⭐⭐⭐

Measurement of central Tendency:

(1) mean
→ Goal is to find center of data

Marks = [10, 20, 30]
↑ ↑ ↑
s1 s2 s3

→ $\frac{\text{mean}}{\text{average}} = \frac{\sum x_i}{n} = \frac{10+20+30}{3}$

? Average marks of my class

$= \frac{60}{3} = 20$

min ✗ max ✗

10 40 30 20

→ 10 20 30 40
↓
avg → mean

? → average → single no
→ 50

Aggregate data → summary

A → 60 → okay
B → 80 → Best
C → 40 → poor
↑
mean

Drawback:

~~aggregated → min & max~~

→ sensitive to outliers

A → [10 , 20 , 40 , 50] → $120/4$ → $\boxed{30}$

B → [10, 20 , 40 , 90] → $160/4$ → 40

C → [10, 20, 40, 60] → $130/4$ → 32.5

D → [10, 20, 40, 500] → 142

500 → outlier

→ mean shifted by a large number bcoz of 500

A → 60 mean → A is better

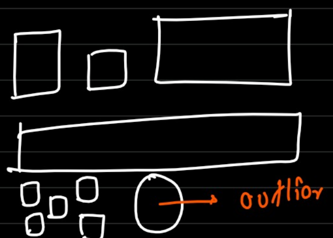
B → 40 ? A has better performing students

↓

40, 40, 40, 40 → 1 student → 100

→ 10, 10, 20

outliers :



Dog_1.jpg Dog_2.jpg
cat_1.jpg cat_2.jpg
lion_1.jpg → outlier

(2) Median (Middle value)

A → 60, 50, 40, 30, 10, 50, 40, 60, 55

sort(A) → 10, 30, 40, 40, 50, 50, 55, 60, 60 → len() → 9

median = 50

len → 9 values

even → 2 centre value → avg → median

odd → take the no. → median

sort(A) → 10, 30, 40, 40, 50 | 50, 55, 60, 60, 80

value → $\frac{50+50}{2}$ → 50 → median

sort(A) → 10, 30, 40, 40, 50, 50, 55, 60, 60, 8000

50

if i want a aggregation that shows the behavior of my data → mean / median ✓

if i want a true center of my data → mean/median ✓

(3) Mode → categorical

Sports → A, B, C, D

1 A
2 B
3 A
4 B
5 A
6 B
7 C
8 C
9 C
10 C

→ mean → ? ✗
→ median → ?

? who won the most game

frequency of categories:

A : 3
B : 3
C : 4

Marks → 1 – 10

60 students →

A	B	C	D	F
1	2	8	7	8

↓
8

Most frequent value → numerical → categorical data
Most frequent value → categories → categorical data

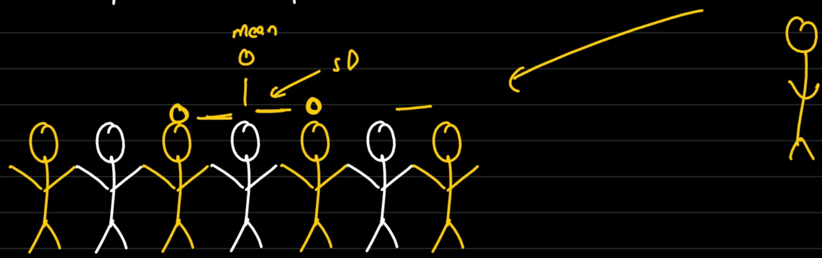
Q. [10, 20, 30, 1000]

centre :— median

mean → ? → sensitive to outliers

Measurement of Dispersion (spread of data)

spread = 2 hands
standard deviation



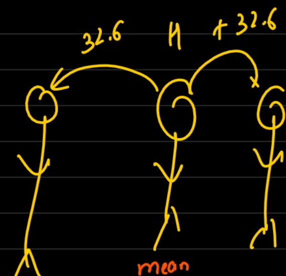
→ what is the common spread between multiple people head?

→ 2 hands

	A	B	C	D	E
Height (cm) →	170	180	100	140	190

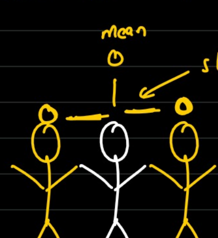
→ 32.6
↓
common spread

→ on a average difference between people height is 32.6



Standard Deviation

SD : How far each point is from mean



A → 50, 51, 49, 50, 52 → 50.4 (mean)

B → 0, 100, 10, 90, 50 → 50 (mean)

consistent
unpredictable → mean, median, mode

→ unstable

base performance → differ → ↑↓ → stable

↓

standard deviation

↓

base

↓

average → mean

series → [10, 20, 30]

→ mean → $10+20+30/3 \rightarrow 20$

→ variance → $\sum \frac{(x_i-\mu)^2}{n}$ → σ^2 (lower case sigma)

avg of (diff of each point from mean)^2

how far? in what direction? ✗

↓

$(10-20)^2 \rightarrow (-10)^2 = 100$

$(20-20)^2 \rightarrow (0)^2 = 0$ → $\frac{100+0+100}{3} = 66.67 \text{ unit}^2$

$(30-20)^2 \rightarrow (10)^2 = 100$

$\approx$ → approx $\pi = 3.14$ ①

$=$ → equal $\pi \approx 3.14$ ②

Standard Deviation → root of variance => $\sigma^2 \approx 66.67 \rightarrow \sqrt{66.67}$

$\sigma \approx 8.16$

$\boxed{8.16}$

66.67 unit^2

$\sigma^2 \rightarrow \sigma$ (goal)

$\sigma^2 \rightarrow \sqrt{\sigma^2} \rightarrow \sigma$

↑

procedure

[10, 20, 30]   SD: 8.16

Series :

on average, numbers deviate from mean by 8.16

average spread → SD

[10   20   30]

$- 8.16$ $+ 8.16$

[11.84   20   28.16]

10 (inch)   20 (inch)   30 (inch)

variance → x inch^2 | x inch

① ②

same unit as the data

SD → x $\boxed{\text{inch}}$